Project Cost Management

by:
Benilda Eleonor V. Comendador, DIT

Project Cost Management includes the processes involved in planning, estimating, budgeting, financing, funding,
managing, and controlling costs so that the project can be completed within the approved budget.

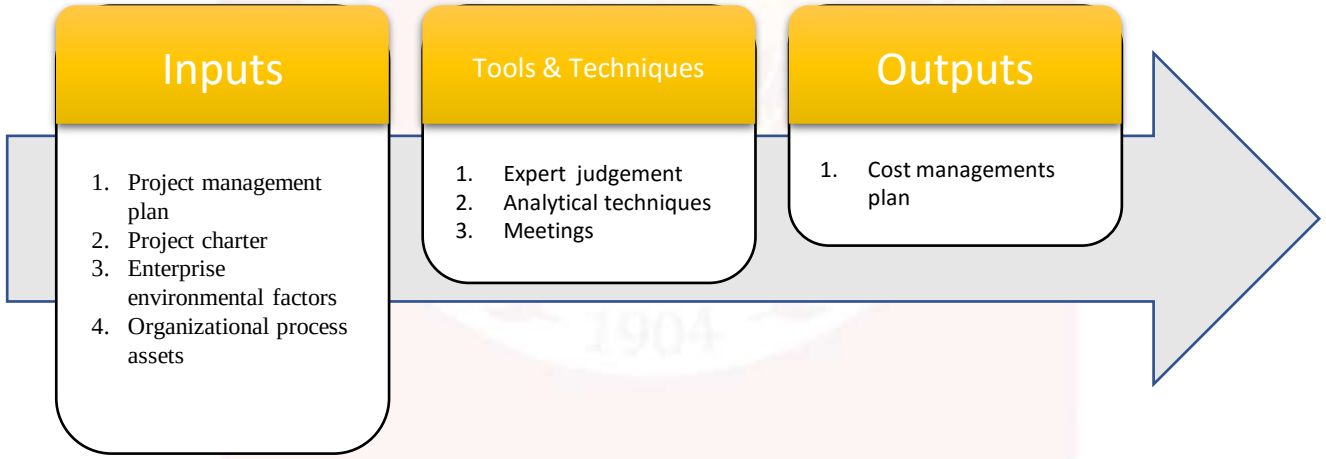
Overview of the Project Cost Management

- ✓ Plan Cost Management
- ✓ Estimates Costs
- ✓ Determine Budget
- ✓ Control Costs

✓Plan Cost Management

Plan Cost Management is the process that establishes the policies, procedures, and documentation for planning, managing, expending, and controlling project costs.

Plan Cost Management



Plan cost Management: Inputs, tools & techniques, and outputs

✓Estimate Costs

Estimate costs is the process developing an approximation of the monetary resources needed to complete project activities.

Estimate Costs

Inputs	Tools & Techniques	Outputs
1. Costs management plan 2. Human resource management plan 3. Scope baseline 4. Project schedule 5. Risk register 6. Enterprise environmental factors 7. Organizational process assets	1. Expert judgement 2. Analogous estimating 3. Parametric estimating 4. Bottom-up estimating 5. Three-point estimating 6. Reserve analysis 7. Costs of quality 8. Project management soft	1. Activity cost estimates 2. Basis of estimates 3. Project documents updates

Estimate costs: Inputs, tools & techniques, and outputs

✓Determine Budget

Determine Budget is the process of aggregating the estimated costs of individual activities or work packages to establish an authorized cost baseline.

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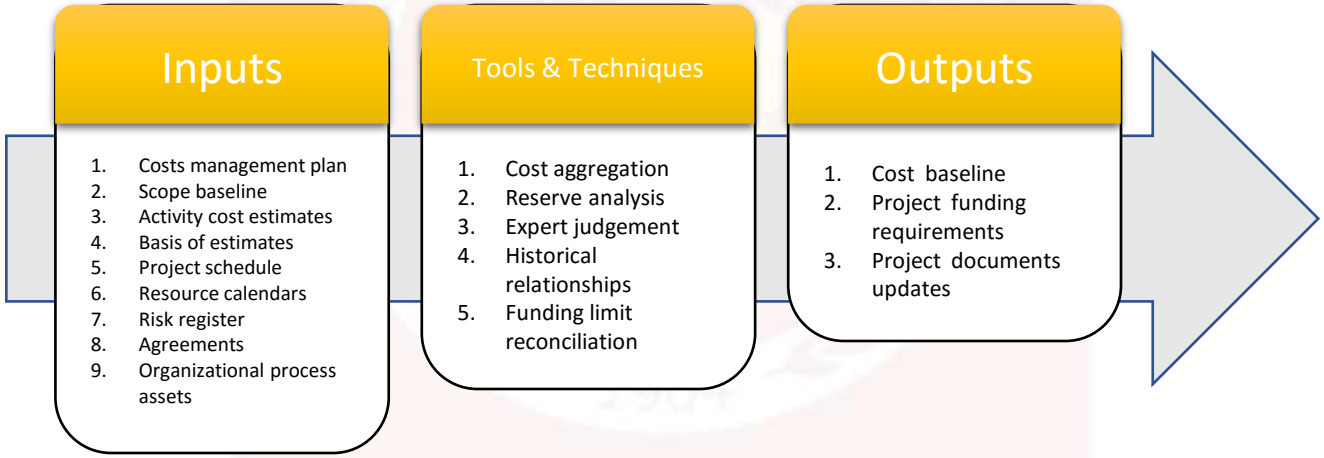
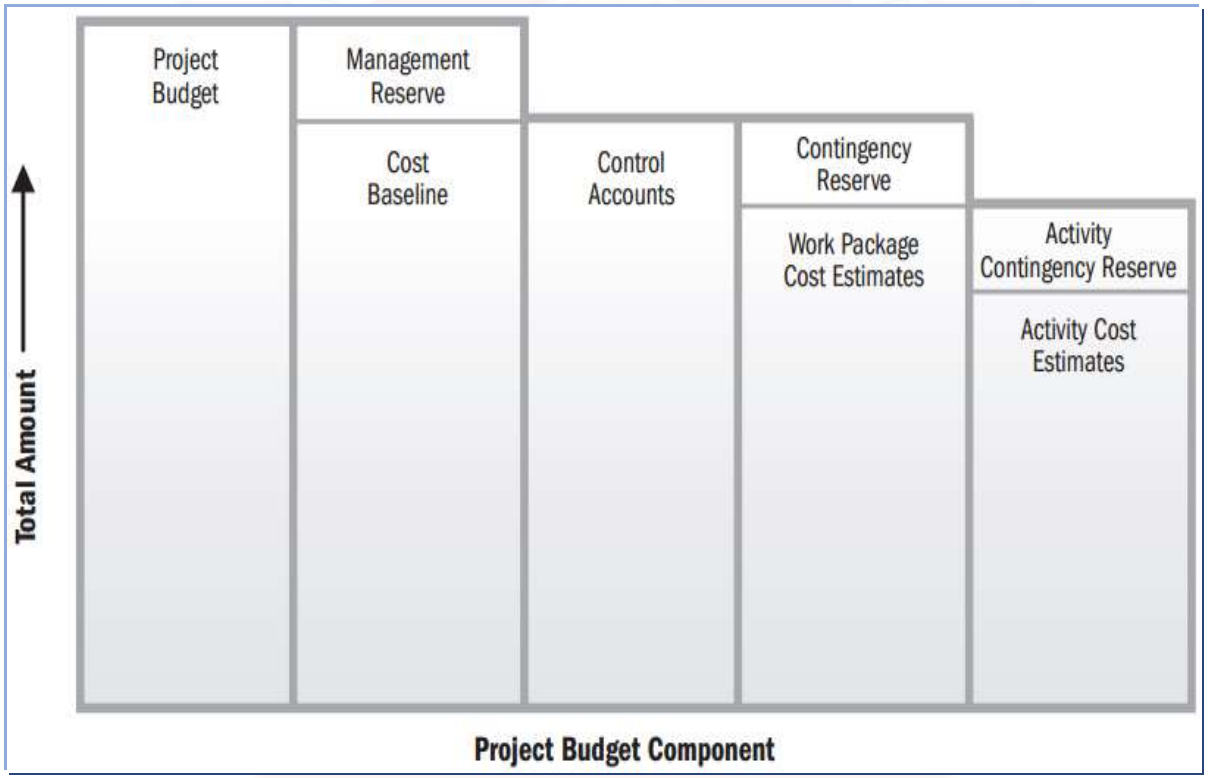
Determine Budget

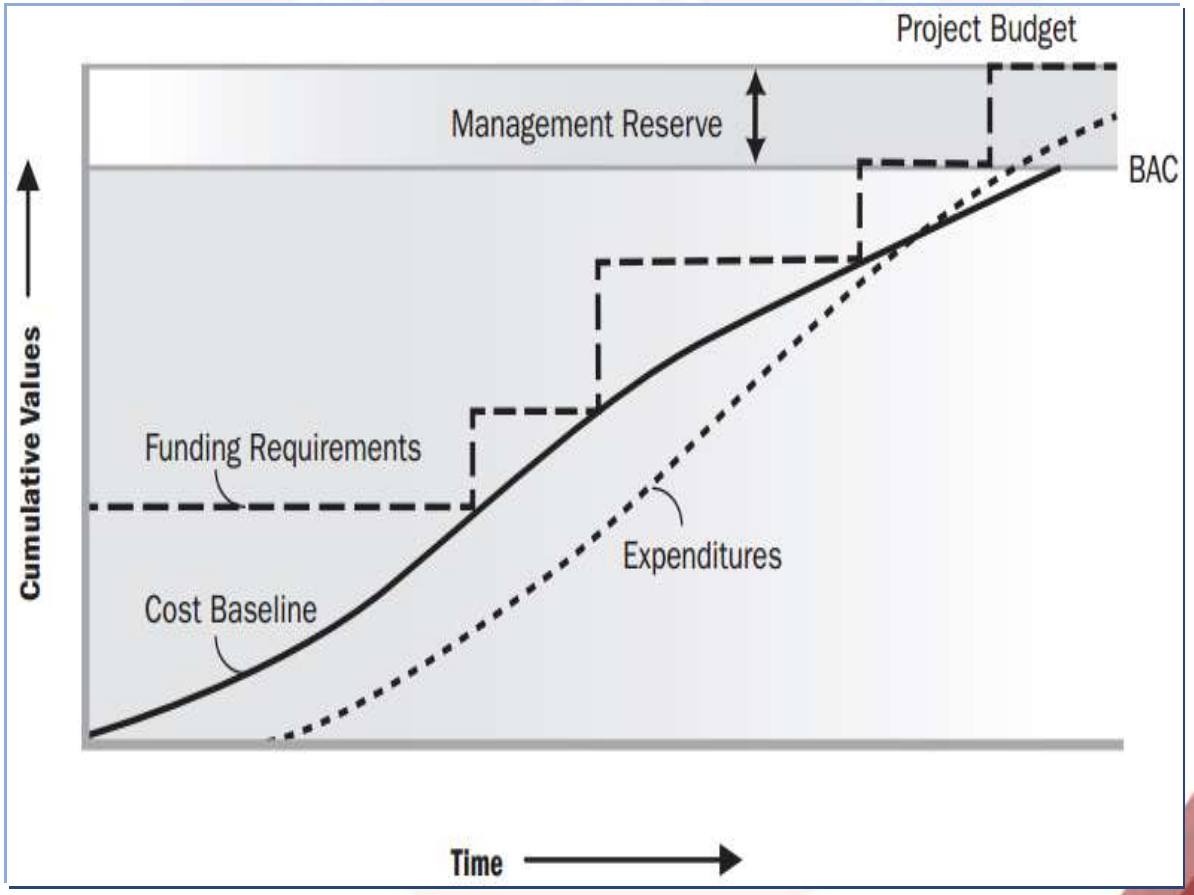


Determine Budget: Inputs, tools & techniques, and outputs

Project Budget components



Cost Baseline, Expenditures, and Funding requirements



✓Control Costs

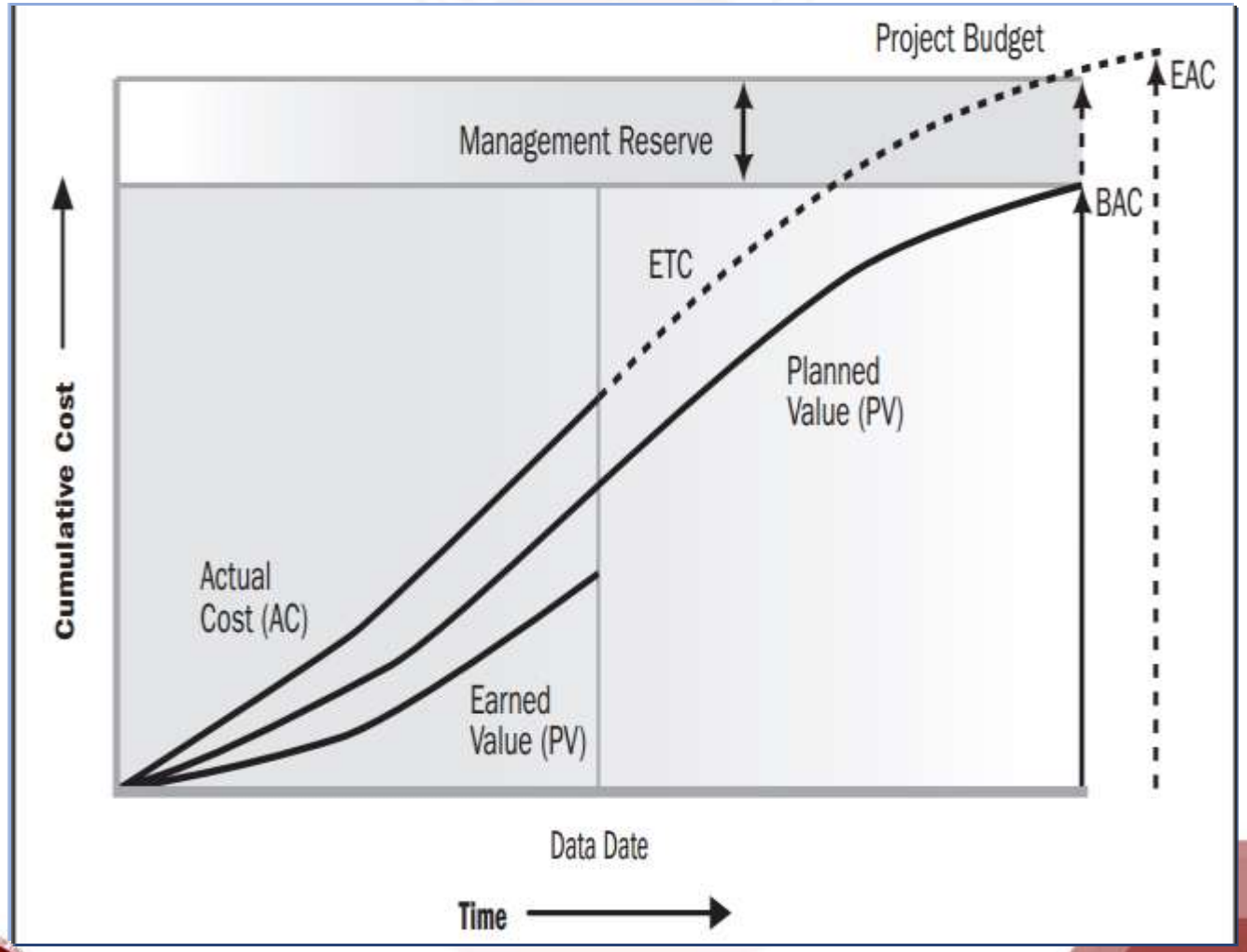
Control Costs is the process of monitoring the status of the project to update the project costs and managing changes to the cost baseline.

Control Cost

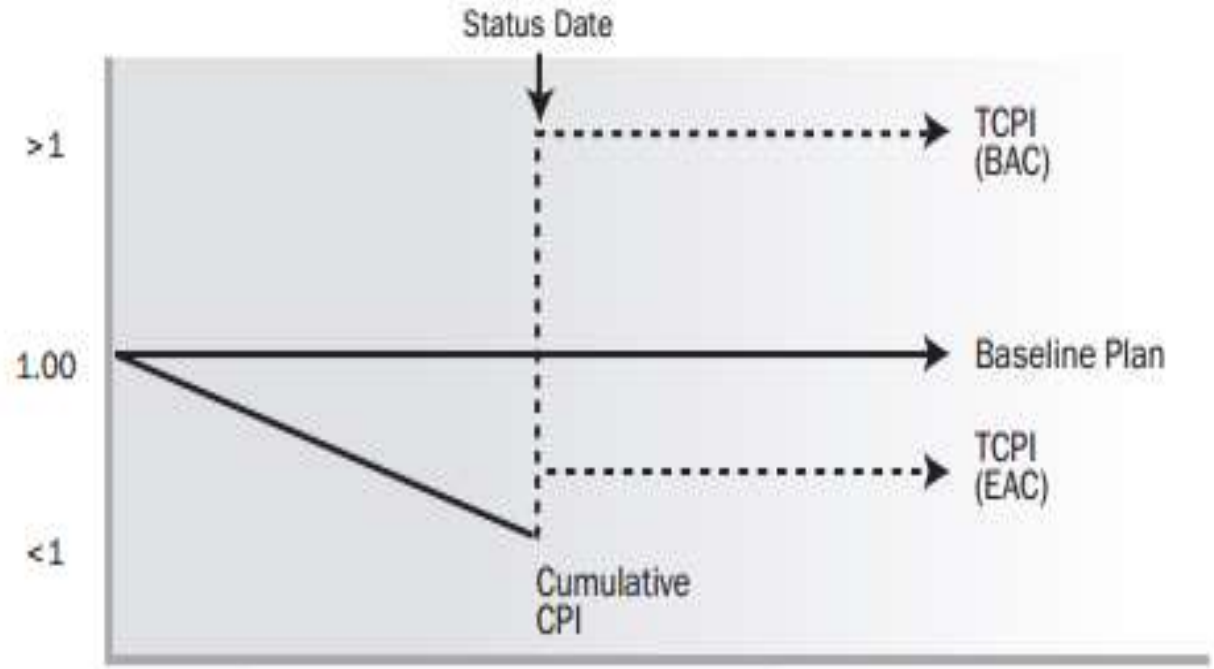
Inputs	Tools & Techniques	Outputs
1. Project management plan 2. Project funding requirements 3. Work performance data 4. Organizational process assets	1. Earned value management 2. Forecasting 3. To-complete performance index(TCPI) 4. Performance review 5. Project management software 6. Reserve analysis	1. Work performance information 2. Cost forecasts 3. Change request 4. Project management plan updates 5. Project documents updates 6. Organizational process assets updates

Control Costs: Inputs, tools & techniques, and outputs

Earned Value, Planned Value, and Actual costs



To-Complete Performance Index (TCPI)



Formula:

$$\frac{\text{Work Remaining (BAC-EV)}}{\text{Funds Remaining (BAC-AC) or (EAC-AC)}} = \text{TCPI}$$

5.3 Cost Estimating

5.3.2 Analogous Estimating (Top Down Method)

- using the actual cost of a previous, similar project as the basis for estimating the cost of the current project.

- It is less costly than other techniques, but it is also less accurate.

5.3 Cost Estimating

- It is most reliable when the previous projects are similar in fact, and the estimators have efficient expertise.

5.3 Cost Estimating

5.3.3 Parametric Modeling

- Using project characteristics in a mathematical model to predict project costs.
- Models may be simple or complex, e.g. one model of software development.
- Both the cost and accuracy of parametric models vary widely.

5.3.3 Parametric Modeling

They are most likely to be reliable when

- the historical information used to develop the model was accurate
- the parameters used in the model are readily quantifiable

5.3.4 Bottom-Up Estimating

- Estimating the cost of individual activities, then summarizing the individual estimates to get a project total.
- Smaller activities increase both cost and accuracy of the estimating process.

5.3.5 Computerized Tools

- Improving cost accumulating efficiency using PMS (Project Management Software) or spreadsheet programs.
- Previously-stated 3 techniques have been easier to be utilized.

Cost Budgeting

Allocating the Overall Cost Estimates to Individual Activities

Inputs

- Cost Estimates
- WBS
- Project Schedule
- Risk Management Plan

Tools and Techniques

- Analogous Estimating
- Parametric Modeling
- Bottom-Up Estimating
- Computerized Tools
- Other Cost Estimating Methods

Output

- Cost Baseline

A time-phased budget in order to measure and monitor cost performance on the project.

Sample Form of Development Schedule and Corresponding Man-Hours

Phase	January		February		March		April	
	Function	Man-Hours	Function	Man-Hours	Function	Man-Hours	Function	Man-Hours
Basic Design	10	400	15	600				
Detailed Design			5	200	20	800	5	200
Coding					20	800	20	800
Integration Test							20	320
Total Man-Hours		400		800		1600		1320
Cumulative Total		400		1200		2800		4120

Points to keep in mind when you calculate man-hours

- Difficulty level of the Project
  Precedents, the level of specification definition, complexity, expertise

5.4 Cost Budgeting

- Skill level of the members of development team
  OS, DB, Network, Tool, Business Knowledge
- Development Environment
  Development System, Development Machine, Development Tool

5.4 Cost Budgeting

5.5.1 What is Cost Control?

- Implement cost change to get good results.
- Decide budget change.
- Do change control when cost change has been implemented.

Cost Control includes:

5.4 Cost Budgeting

5.5.1 What is Cost Control?

- Monitoring cost performance to detect variances from plan.
- Ensuring that all appropriate changes are accurately built in.
- Preventing incorrect, inappropriate, or unauthorized changes from being built in.
- Informing appropriate stakeholders of authorized changes.

5.5 Cost Control

Inputs

- Cost Baseline
- Performance Report
- Change Requests
- Cost Management Plan

Tools and Techniques

- Cost Change Control System
- Performance Measurement
- Earned Value Management (EVM)
- Additional Planning
- Computerized Tools

Outputs

- Revised Cost Estimates
- Budget Updates
- Corrective Action
- Estimate at Completion
- Project Closeout
- Lessons Learned

Thank you for your Attention !

Q and A

Project Quality Management

by:
Benilda Eleonor V. Comendador, DIT

Overview of Project Quality Management

- ✓ Plan Quality Management
- ✓ Perform Quality Assurance
- ✓ Control Quality

Project Quality Management

- the processes and activities of the performing organization that determine quality policies, objectives, and responsibilities.

- **Plan Quality Management**
  - identifying and documenting project roles, responsibilities, and reporting relationships.

- **Perform Quality Assurance**

  - auditing the quality requirements and the results from quality control measurements

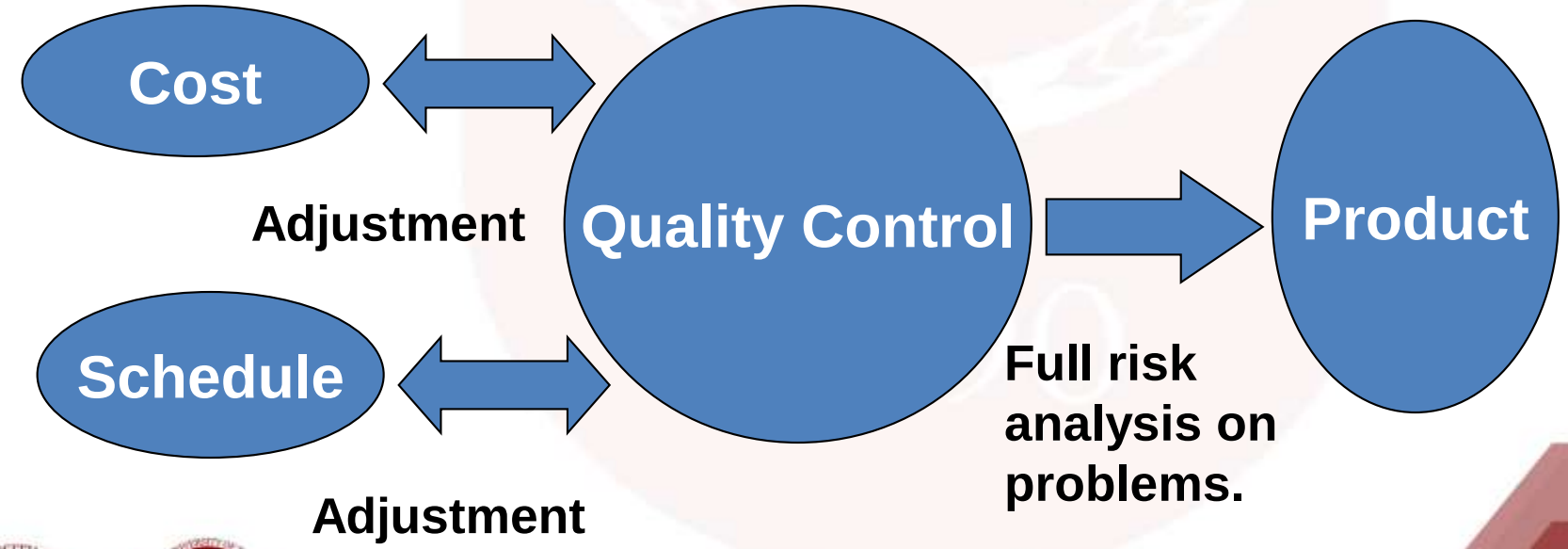
- **Control Quality**
  - Monitoring and recording results of executing the quality activities to assess performance.

Plan Quality Management: Inputs, Tools & Techniques, and Outputs

Inputs	Tools & Techniques	Outputs
.1 Project management plan .2 Stakeholder register .3 Risk register .4 Requirements documentation .5 Enterprise environmental factors .6 Organizational process assets	.1Cost-benefit analysis .2 Cost of quality .3 Seven basic quality tools .4 Benchmarking .5 Design of experiments .6 Statistical sampling .7 Additional quality planning tools meetings	.1 Quality management plan .2 Process improvement plan .3 Quality metrics .4 Quality checklists .5 Project documents updates

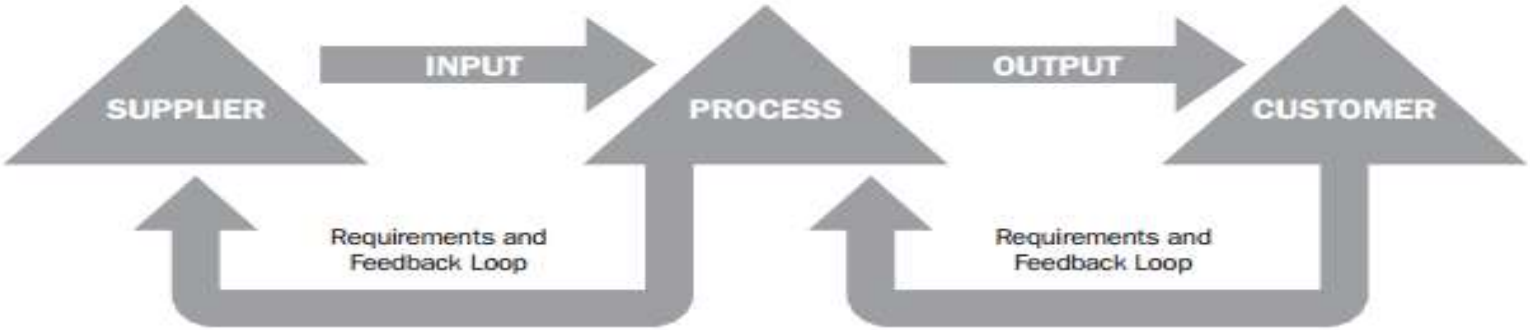
Quality Planning

- Performing in parallel with the other planning processes



SIPOC Model

Suppliers	Inputs	Process	Outputs	Customers
•	•	•	•	•
•	•	•	•	•
•	•	•	•	•
•	•	•	•	•



Requirements List	Measurements List	Requirements List	Measurements List
•	•	•	•
•	•	•	•
•	•	•	•
•	•	•	•

NOTE: The components of this diagram are flexible and can take any direction depending upon the circumstance.

Perform Quality Assurance: Inputs, Tools & Techniques, and Outputs

Inputs

1. Quality management plan
2. Process improvement plan
3. Quality metrics
4. Quality control measurements
5. Project documents

Tools & Techniques

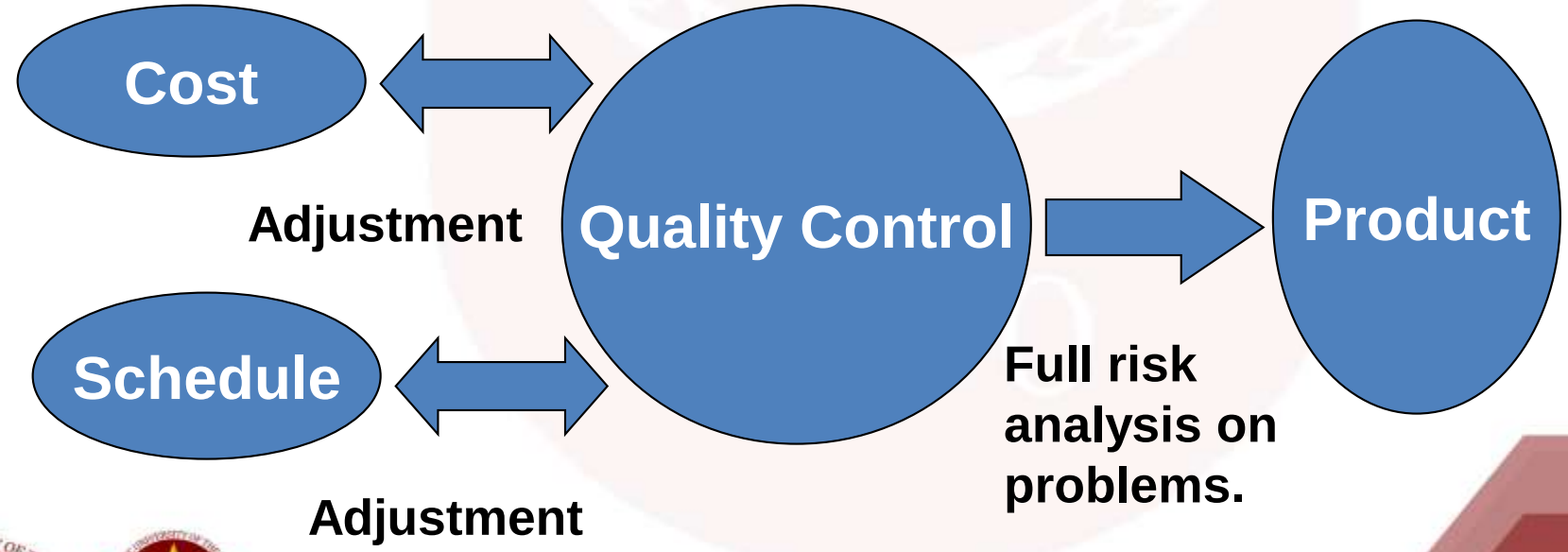
1.Quality management and control tools
2. Quality audits analysis

Outputs

1. Change requests
2. Project management plan updates
3. Project documents updates
4. Organizational process assets updates

Quality Planning

- Performing in parallel with the other planning processes



Control Quality: Inputs, Tools & Techniques, and Outputs

Inputs

1. Project management plan
2. Quality metrics
3. Quality checklists
4. Work performance data
5. Approved change requests
6. Deliverables
7. Project documents
8. Organizational process assets

Tools & Techniques

1. Seven basic quality tools
2. Statistical sampling
3. Inspection
4. Approved change requests review

Outputs

1. Quality control measurements
2. Validated changes
3. Verified deliverables
4. Work performance information
5. Change requests
6. Project management plan updates
7. Project documents updates
8. Organizational process assets updates

Thank you for your Attention !

Q and A

Project Human Resource Management

by:
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Project Human Resource Management

- the processes that organize, manage, and lead the project team.

Project Human Resource Management Processes

- **Plan Human Resource Management**

  - identifying and documenting project roles, responsibilities, and reporting relationships.

- **Acquire Project Team**

  - confirming human resource availability and obtaining the team necessary to complete project activities.

- **Develop Project Team**

  - improving competencies, team member interaction, and overall team environment to enhance project performance.

- **Manage Project Team**

  - Tracking team member performance, providing feedback, resolving issues, and managing changes.

Project Human Resource Management Overview

Plan Human Resource Management

.1 Input
- .1 Project management plan
- .2 Activity resource requirements
- .3 Enterprise environmental factors
- .4 Organizational process assets

.2 Tools & Techniques
- .1 Organizational charts and position descriptions
- .2 Networking
- .3 Organizational theory
- .4 Expert judgment
- .5 Meetings

.3 Outputs
- .1 Human resource management plan

Acquire Project Team

.1 Input
- .1 Human resource management plan
- .2 Enterprise environmental factors
- .3 Organizational process assets

.2 Tools & Techniques
- .1 Pre-assignment
- .2 Negotiation
- .3 Acquisition
- .4 Virtual teams
- .5 Multi-criteria decision analysis

.3 Outputs
- .1 Project staff assignments
- .2 Resource calendars
- .3 Project management plan updates

Develop Project Team

.1 Input
- .1 Human resource management plan
- .2 Project staff assignments
- .3 Resource calendars

.2 Tools & Techniques
- .1 Interpersonal skills
- .2 Training
- .3 Team-building activities
- .4 Ground rules
- .5 Colocation
- .6 Recognition and rewards
- .7 Personnel assessment tools

.3 Outputs
- .1 Team performance assessments
- .2 Enterprise environmental factors updates

Manage Project Team

.1 Input
- .1 Human resource management
- .2 Project staff assignments
- .3 Team performance assessments
- .4 Issue log
- .5 Work performance reports
- .6 Organizational process assets

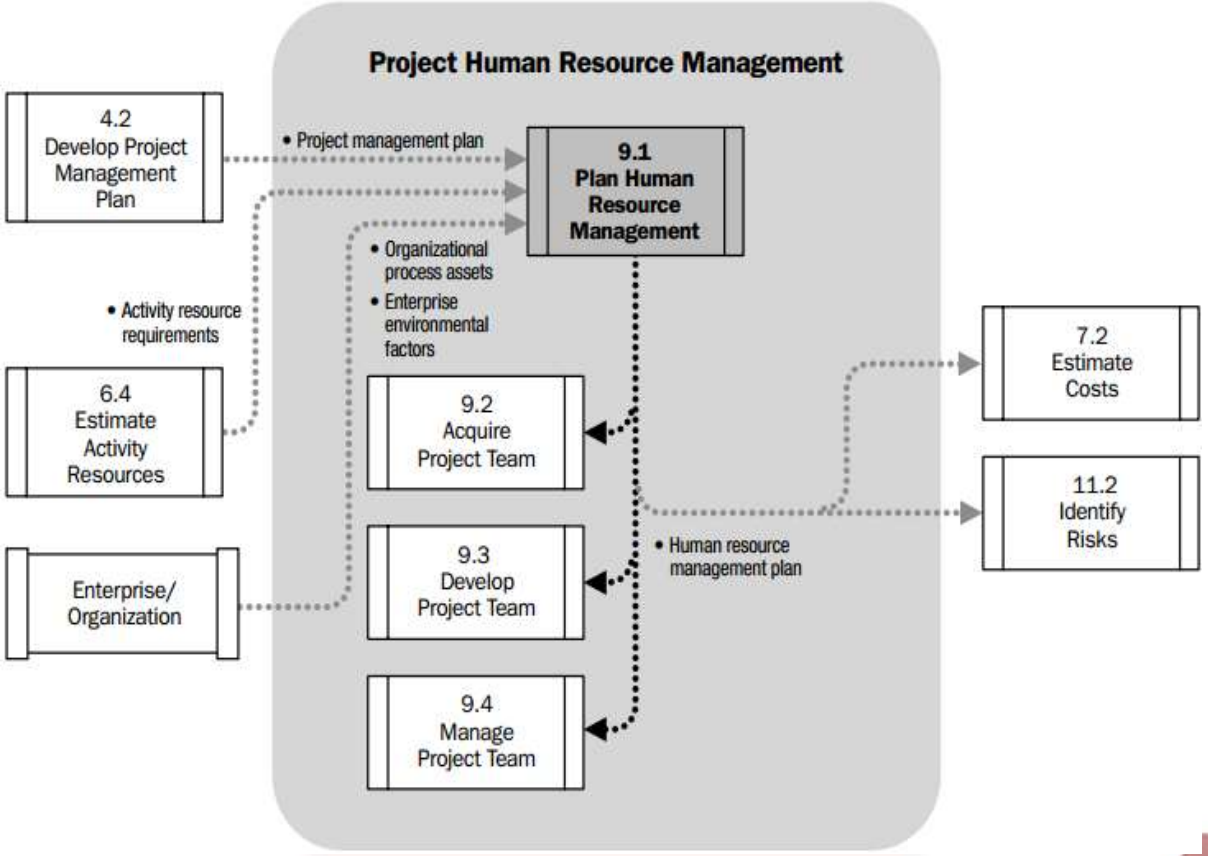
.2 Tools & Techniques
- .1 Observation and conversation
- .2 Project performance appraisals
- .3 Conflict management
- .4 Interpersonal skills

.3 Outputs
- .1 Change requests
- .2 Project management plan updates
- .3 Project documents updates
- .4 Enterprise environmental factors updates
- .5 Organizational process assets updates

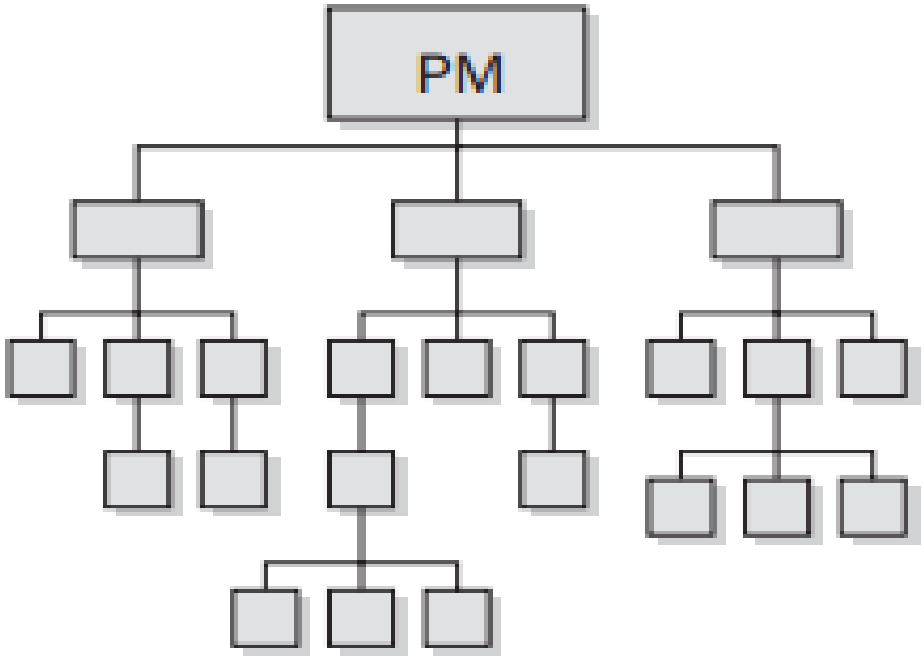
Plan Human Resource Management: Inputs, Tools & Techniques, and Outputs

Inputs	Tools & Techniques	Outputs
.1 Project management plan .2 Activity resource requirements .3 Enterprise environmental factors .4 Organizational process assets	.1 Organizational charts and position descriptions .2 Networking .3 Organizational theory .4 Expert judgment .5 Meetings	.1 Human resource management plan

Plan Human Resource Management Data Flow Diagram



Roles and Responsibility Definition Formats



Organization Chart (hierarchical)

Roles and Responsibility Definition Formats

- Hierarchical-type chart

  - The traditional organizational chart structure can be used to show positions and relationships in a graphical, top-down format.

Roles and Responsibility Definition Formats

RAM				

Responsibility Chart (matrix)

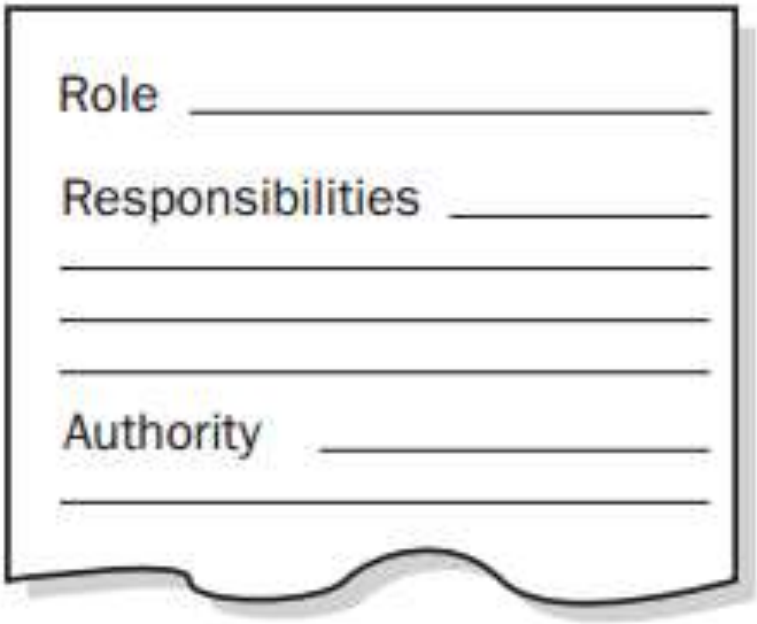
Roles and Responsibility Definition Formats

- Matrix-based charts

  - A responsibility assignment matrix is a grid that shows the project resources assigned to each work package.

Ex. RACI (responsible, accountable, consult, and inform)

Roles and Responsibility Definition Formats



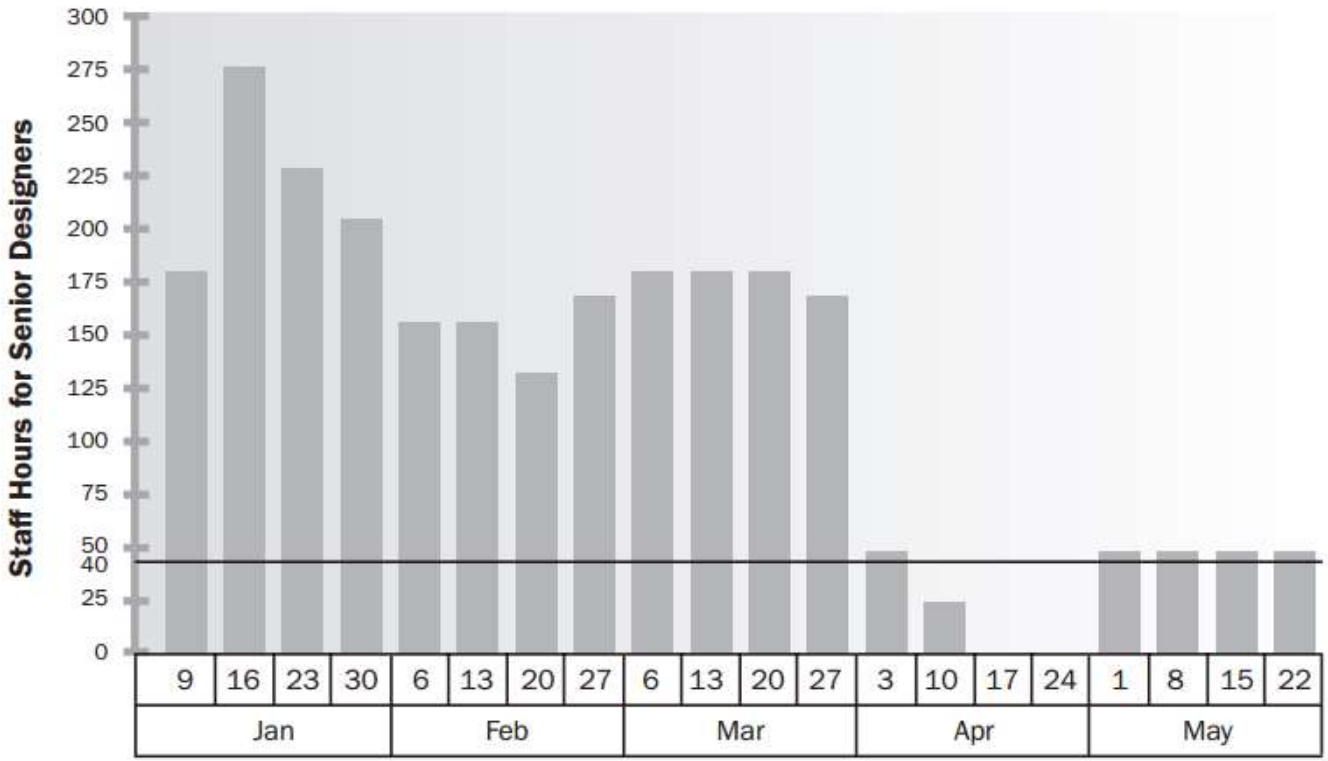
Role Description (text)

Roles and Responsibility Definition Formats

- Text-oriented formats

  - Team member responsibilities that require detailed descriptions can be specified in text-oriented formats.

Staffing Management Plan

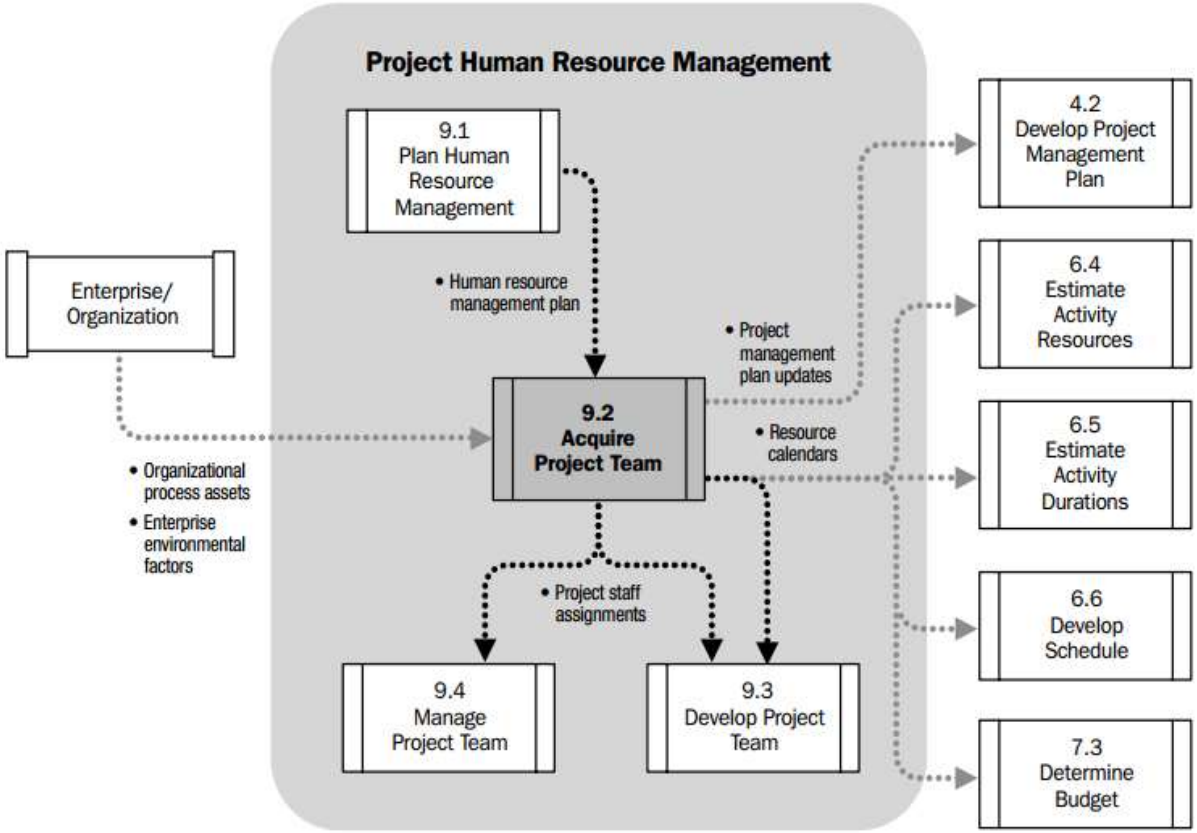


Resource Histogram

Acquire Project Team: Inputs, Tools & Techniques, and Outputs

Inputs	Tools & Techniques	Outputs
.1 Human resource management plan .2 Enterprise environmental factors .3 Organizational process assets	.1 Pre-assignment .2 Negotiation .3 Acquisition .4 Virtual teams .5 Multi-criteria decision analysis	.1 Project staff assignments .2 Resource calendars .3 Project management plan updates

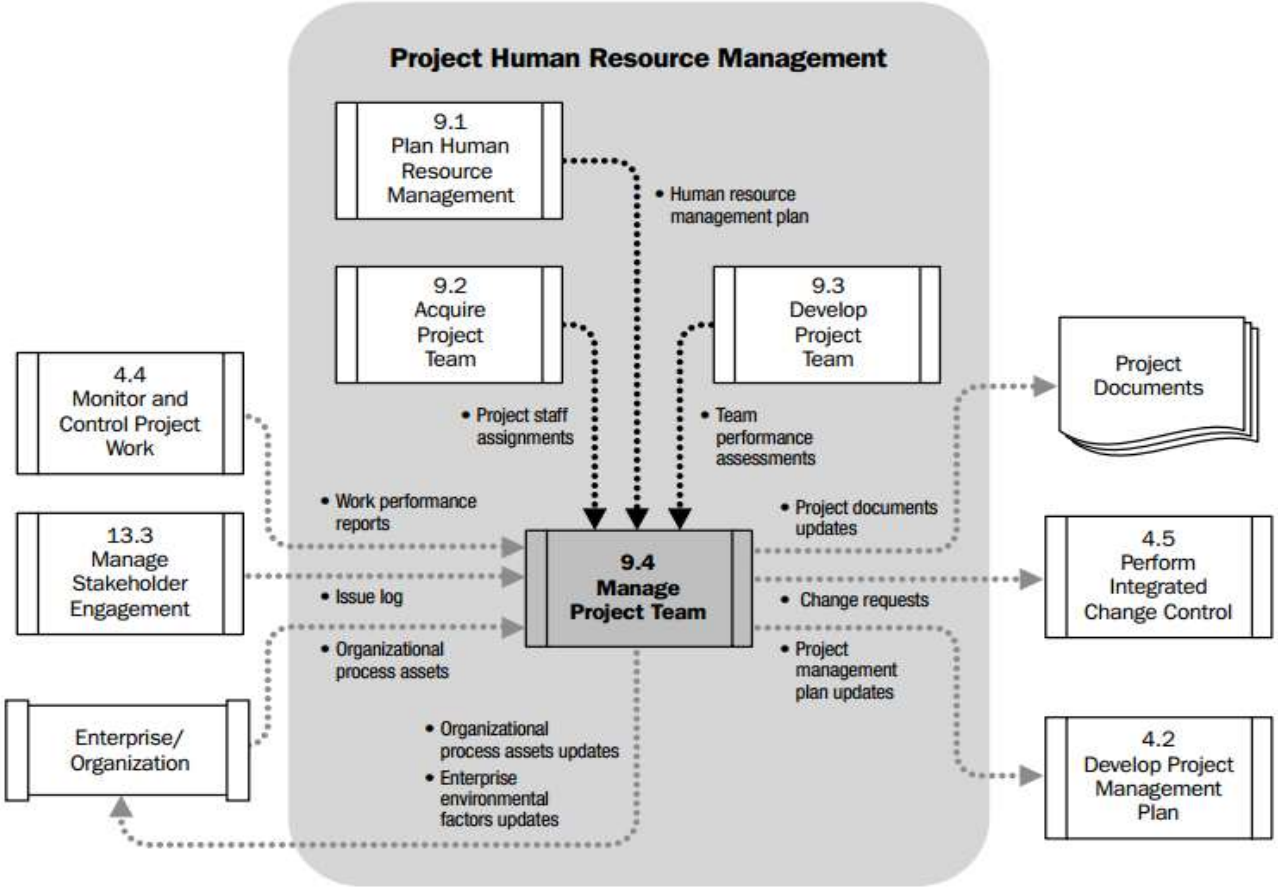
Acquire Project Team Data Flow Diagram



Manage Project Team: Inputs, Tools & Techniques, and Outputs

Inputs	Tools & Techniques	Outputs
1 Human resource management plan .2 Project staff assignments .3 Team performance assessments .4 Issue log .5 Work performance reports .6 Organizational process assets	.1 Observation and conversation .2 Project performance appraisal .3 Conflict management .4 Interpersonal skills	.1 Change requests .2 Project management plan updates .3 Project documents updates .4 Enterprise environmental factors updates .5 Organizational process assets updates

Manage Project Team Data Flow Diagram



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Q and A